



Somatic mutations, Mutational Signatures and Clonal Selection (Part1)

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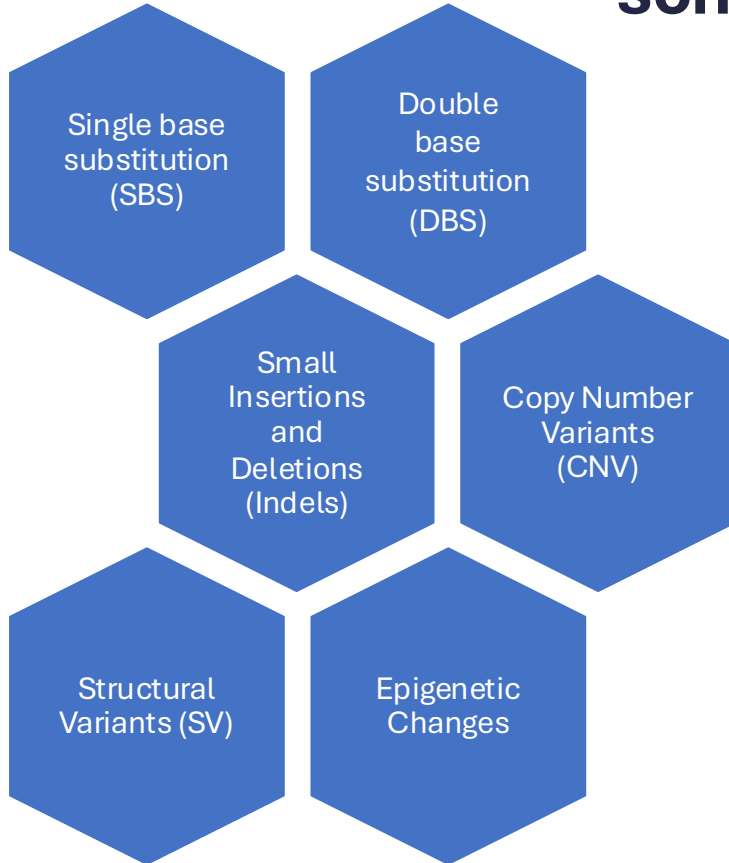
Somatic Genomics Programme

What is a somatic mutation?

A **somatic mutation** is a change in the DNA that occurs in a **non-reproductive (somatic) cell** of the body

- They are not inherited, and only cells descended from the mutated cell will carry the mutation
 - They occur throughout life
- They occur in all tissues (albeit at different rates)

What are the different types of somatic mutation?

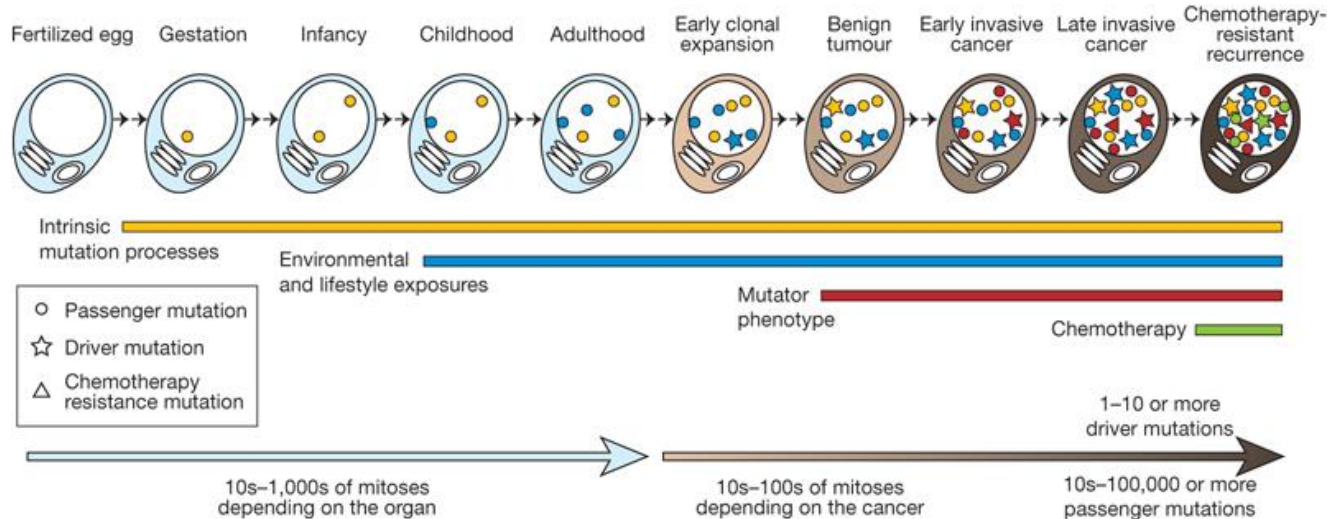


We can classify somatic mutations in many different ways:

- Scale of the DNA change
- Functional impact (Coding / Non-coding)
- Selection advantage (Driver / Passenger)
- Clonal status (Clonal / Subclonal)

Why study somatic mutations?

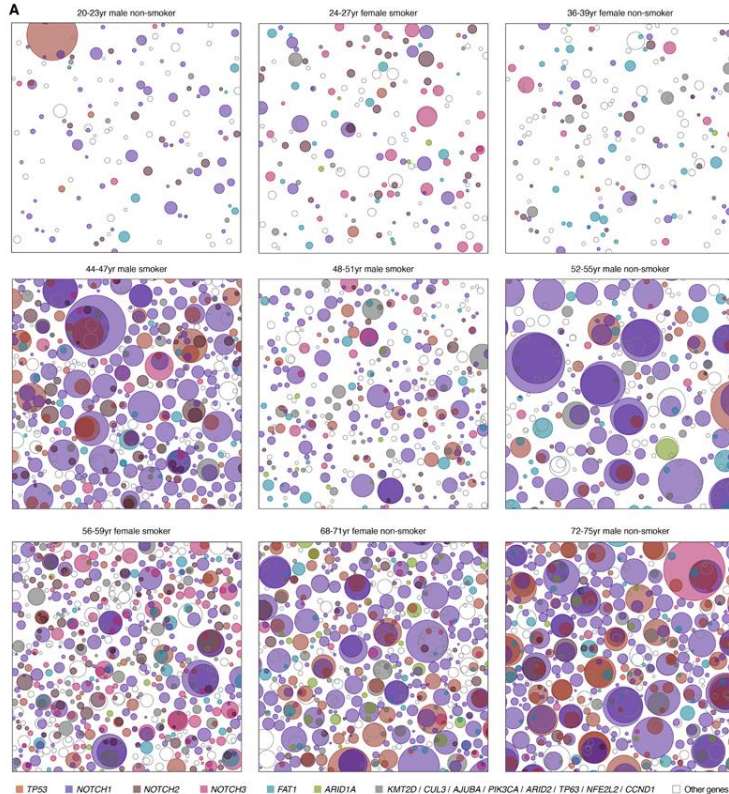
Somatic mutations tell a story of how cells have changed over the course of a lifetime



In cancers we can use somatic mutations as a record of the events which lead to cancer initiation and progression

Figure taken from Stratton et al, The Cancer Genome, Nature, 2009

Why study somatic mutations?



We now know that driver somatic mutations are far more common in normal tissues than was previously thought

Somatic mutations have a role not just in cancer, but also in aging and many other diseases

How do we study somatic mutations?

For a long time, studies were largely restricted to cancers, as ‘bulk’ sequencing is only capable of detecting **clonal** mutations – those which are present in multiple cells.

This is because we are unable to distinguish mutations found in smaller clones/single cells from the background error rate of sequencing technologies

However new technologies have emerged which allow us to more effectively study normal tissues as well

How do we study somatic mutations?

Laser Capture Microscopy + low input DNA sequencing

Allows the capture and sequencing of small clonal structures (for example, colonic crypts)

References:

Ellis et al, Reliable detection of somatic mutations in solid tissues by laser-capture microdissection and low-input DNA sequencing, Nat Protoc 2021

Duplex Sequencing (for example Nanoseq)

Allows the detection of mutations present in a single DNA molecule, allowing the investigation on non-clonal normal tissues

References:

Abascal et al, Somatic mutation landscapes at single-molecule resolution, Nature 2021



How are somatic mutations acquired?

1. Endogenous processes

- Ubiquitous (e.g., DNA replication error, Metabolism, Aging)
- Non-ubiquitous (e.g., APOBEC, DNA repair defects)

2. Exogenous exposures (e.g., UV light and Tobacco)

Each individual mutational process generates different types of mutation
We call these patterns **Mutational Signatures**

What is a Mutational Signature?

C>T
C>A
C>G
T>A
T>C
T>G

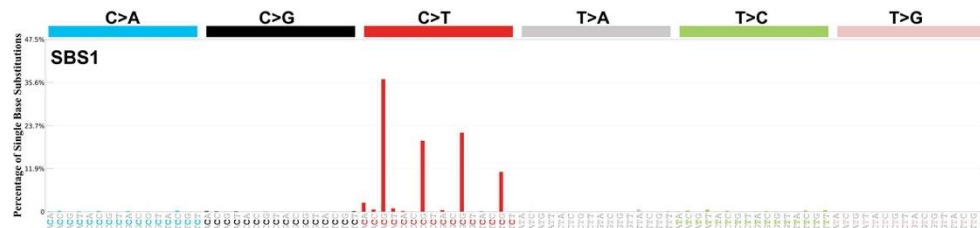
6 mutation classes
(annotated on the
pyrimidine base)

ACA>ATA
ACC>ATC
ACG>ATG
ACT>ATT
CCA>CTA
CCC>CTC
CCG>CTG
CCT>CTT
GCA>GTA
GCC>GTC
GCG>GTC
GCT>GTT
TCA>TTA
TCC>TTC
TCG>TTG
TCT>TTT

.....GGGAAT**C**GGACCCG.....
↓
.....GGGAAT**T**GGACCCG.....
↓
.....GGGAA**T****C**GGACCCG.....
↓
.....GGGAA**T****T**GACCCG.....

16 subtypes of
each mutation
class depending on
the bases on the 5'
and 3' side of the
mutated base

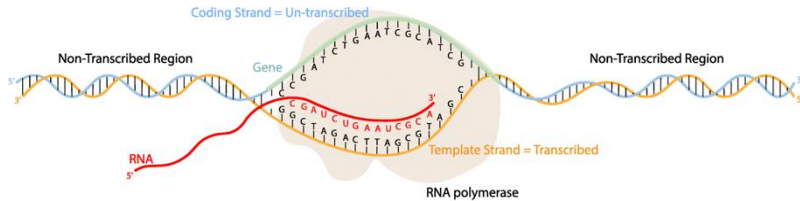
6 mutation classes x 16 subtypes = 96 contexts



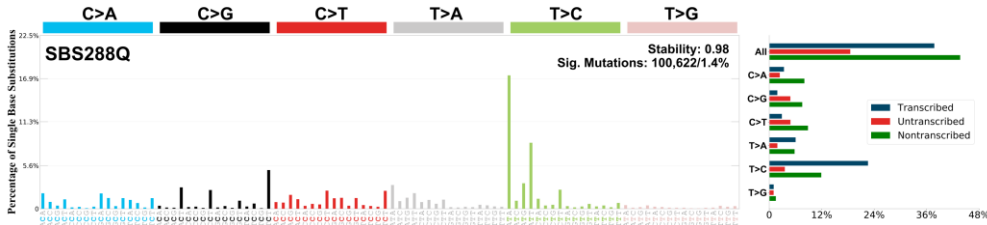
Each bar = the probability of a mutation caused by this biological
process arising in this specific SBS96 context

Extended Contexts

SBS288



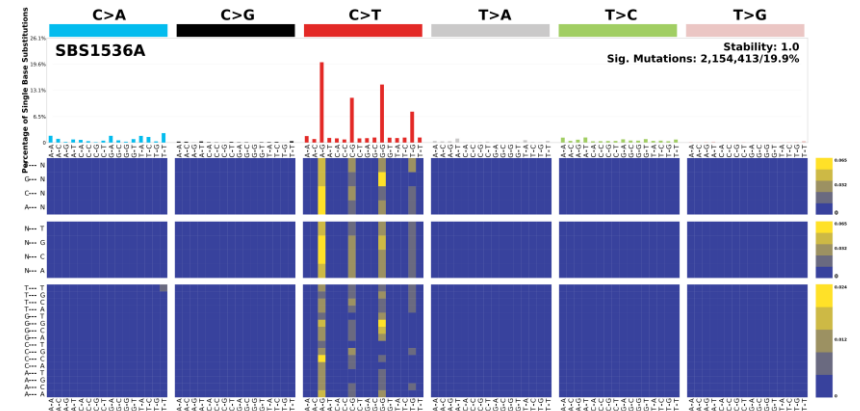
SBS96 x 3 for transcribed, untranscribed and nontranscribed (intergenic) regions



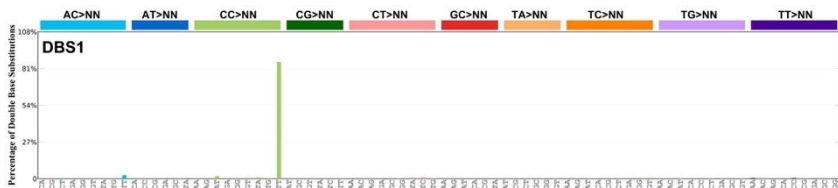
SBS1536

.....GGGA**AT**CGGACCCG.....
 ↓
GGGA**AT**TGGACCCG.....

SBS1536 = pentanucleotide contexts (2 bases either side of the mutated one)



DBS78



DBSs are referred to using the maximum number of pyrimidines; for example, an AA:TT > GT:CA mutation is denoted as TT > AC rather than the alternative AA>GT

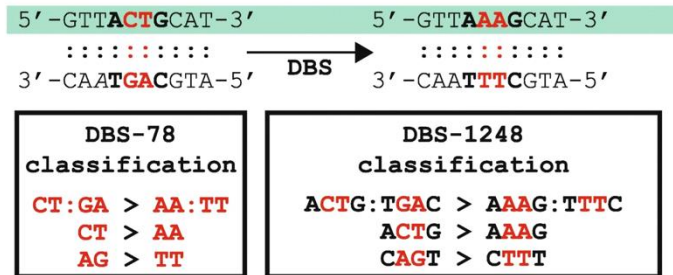
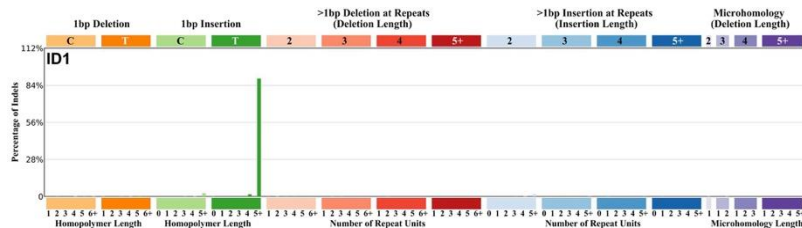


Figure taken from Bergstrom et al, **SigProfilerMatrixGenerator: a tool for visualizing and exploring patterns of small mutational events**, BMC Genomics, 2019

Other Signature Types

ID83



More complex than SBS/DBS

Counts mutations based on:

- Single base pair or longer event
- Length of deletion/insertion
- Repetitive regions or Microhomology

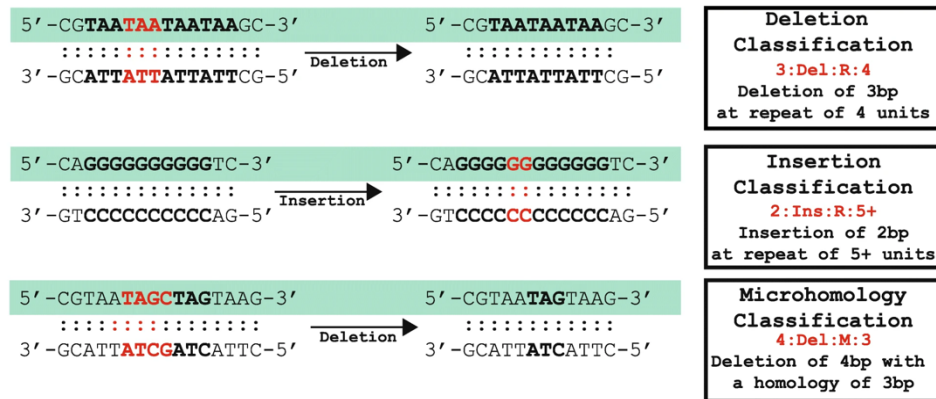
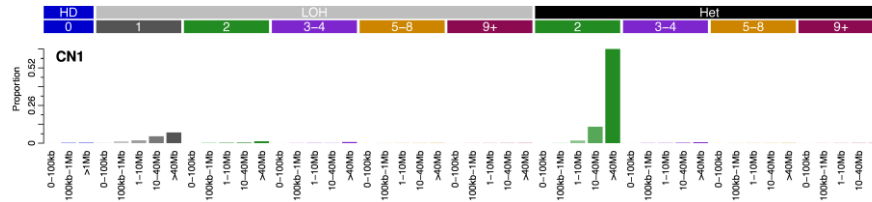


Figure taken from Bergstrom et al, **SigProfilerMatrixGenerator: a tool for visualizing and exploring patterns of small mutational events**, BMC Genomics, 2019

Other Signature Types

CNV48



Counts segments based on:

- Total copy number
- Heterozygosity
- Segment size

Signatures of copy number alterations in human cancer

<https://www.nature.com/articles/s41586-022-04738-6>

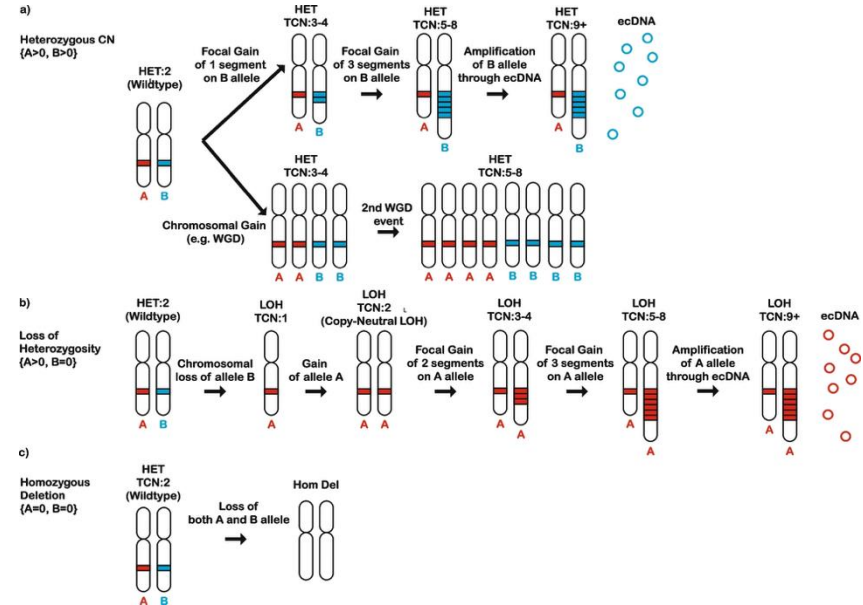
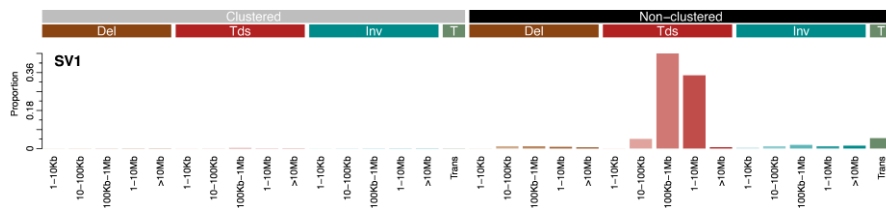


Figure taken from Khandekar et al, **Visualizing and exploring patterns of large mutational events with SigProfilerMatrixGenerator**, BMC Genomics, 2023

Other Signature Types

SV32



Counts variants based on:

- Type (Deletion, Inversion, Tandem Duplication, Translocation)
- Clustered vs Non Clustered
- Size

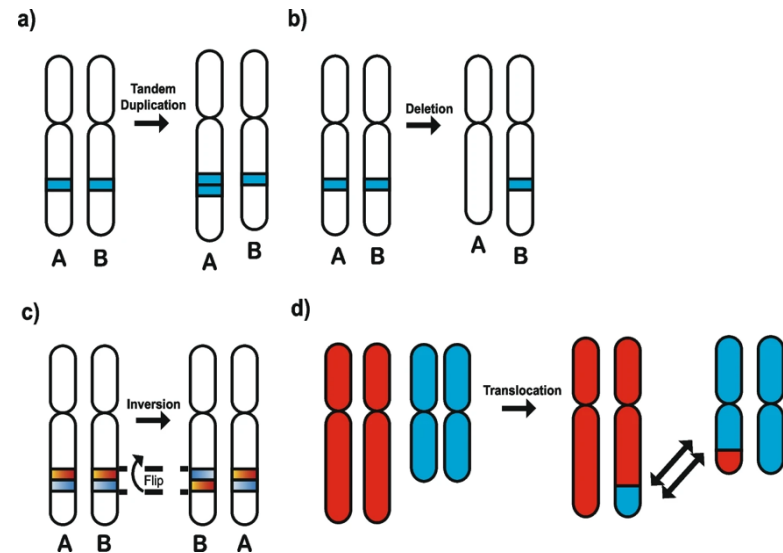


Figure taken from Khandekar et al, **Visualizing and exploring patterns of large mutational events with SigProfilerMatrixGenerator**, BMC Genomics, 2023

Reference Mutational Signatures

Human Cancer Signatures Version 3.5

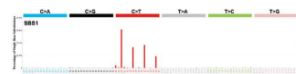
Single Base Substitution (SBS) Signatures

About the collection

Single base substitutions (SBS), also known as single nucleotide variants, are defined as a replacement of a certain nucleotide base. Considering the pyrimidines of the Watson-Crick base pairs, there are only six different possible substitutions: C>A, C>G, C>T, T>A, T>C, and T>G. These SBS classes can be further expanded considering the nucleotide context. Current SBS signatures have been identified using 96 different contexts, considering not only the mutated base, but also the bases immediately 5' and 3'. Click on any signature below to learn more about its details.

Methods

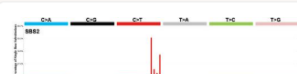
Signature extraction methods With a few exceptions, the current set of reference signatures were extracted using [SigProfiler](#) (as described in [Alexandrov, L.B. et al., 2020](#)) from the 2,780 whole-genome variant calls produced by the ICGC/TCGA [Pan Cancer Analysis of Whole Genomes \(PCAWG\)](#) Network. The stability and reproducibility of the signatures were assessed on somatic mutations from an additional 1,865 whole genomes and 19,184 exomes. All input data and references for original sources are available from [synapse.org ID syn11801889](#). COSMIC mutational signatures are available in numerical form in our [data downloads page](#).



SBS1

Proposed Aetiology

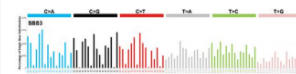
Spontaneous deamination of 5-methylcytosine (clock-like signature)



SBS2

Proposed Aetiology

Activity of APOBEC family of cytidine deaminases



SBS3

Proposed Aetiology

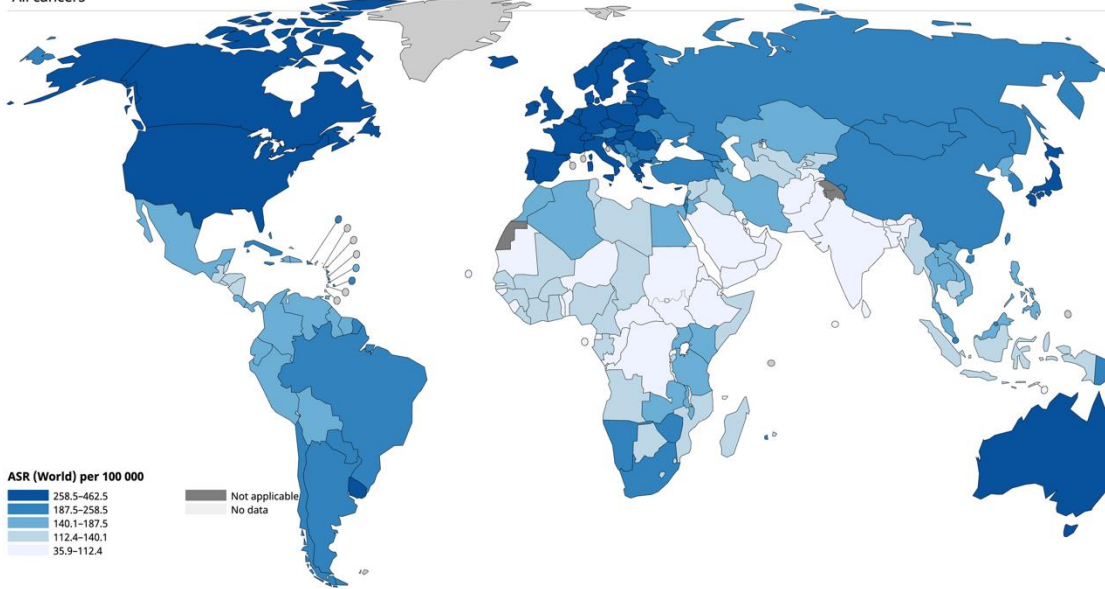
Defective homologous recombination DNA damage repair

The most commonly used reference database for mutational signatures is hosted by COSMIC

<https://cancer.sanger.ac.uk/signatures>

Cancer Incidence

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
All cancers



Cancer incidence rates vary worldwide

In many cases, the reasons are poorly understood but are thought to be due to environment or lifestyle factors which are in theory, preventable

New approaches are needed to complement traditional cancer epidemiology

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Cancer TODAY | IARC
<https://gco.iarc.who.int/today>
Data version: Globocan 2022 (version 1.1) - 08.02.2024
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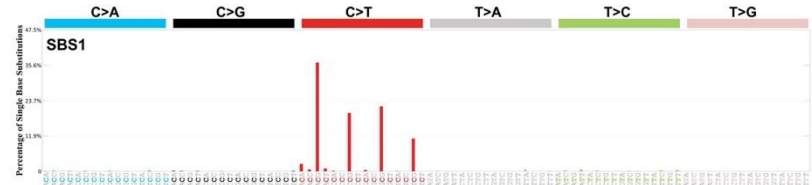
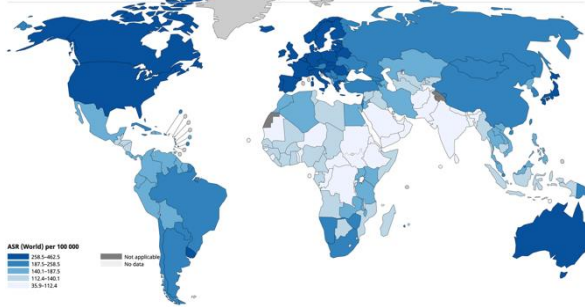
Mutational Epidemiology

The Mutographs Project

Mutagenic carcinogens can be detected via the mutational signatures they imprint upon the genome

By comparing the mutational signature landscape between countries of varying incidence rate and combining this with cancer epidemiology, we may be able to find novel preventable causes of cancer

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
All cancers

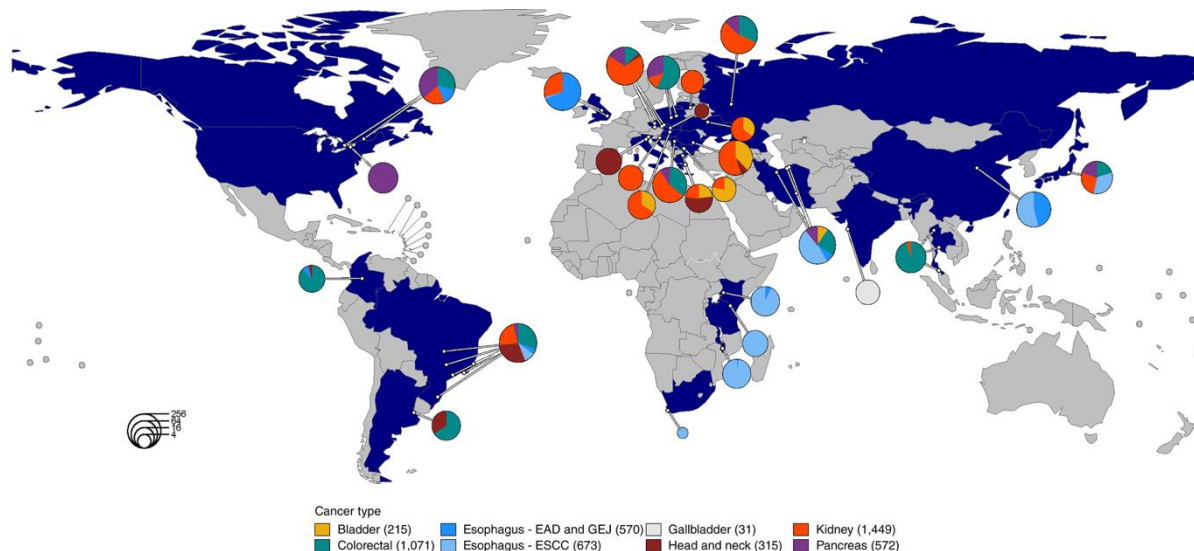


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International Agency
for Research on Cancer
IARC

The Mutographs Project



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Dotted and dashed lines on the map represent approximate border lines for which there may not yet be full agreement.

Data source: Mutographs

Map production: International Agency for Research on Cancer
World Health Organization

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Published Studies:

Esophagus - ESCC

Kidney

Colorectal

Head and Neck

Manuscript Preparation:

Pancreas

Esophagus - EAD/GEJ

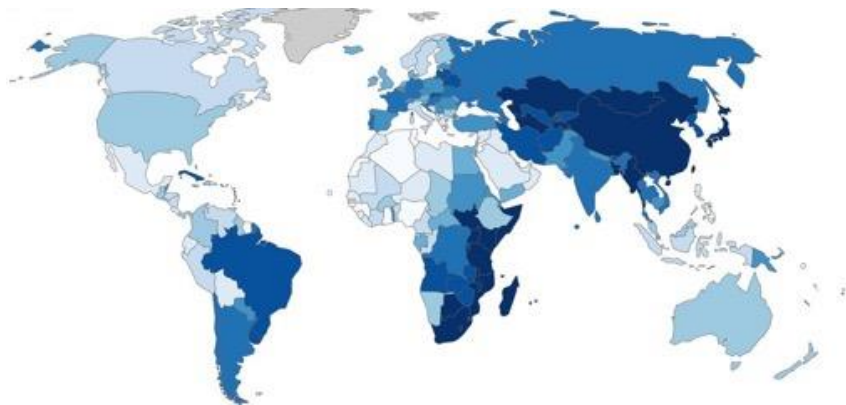
Data Collection:

Bladder

In total, over 15,000 genomes sequenced across 30 countries in 5 continents

Esophageal Squamous Cell Carcinoma (ESCC)

ESCC Incidence

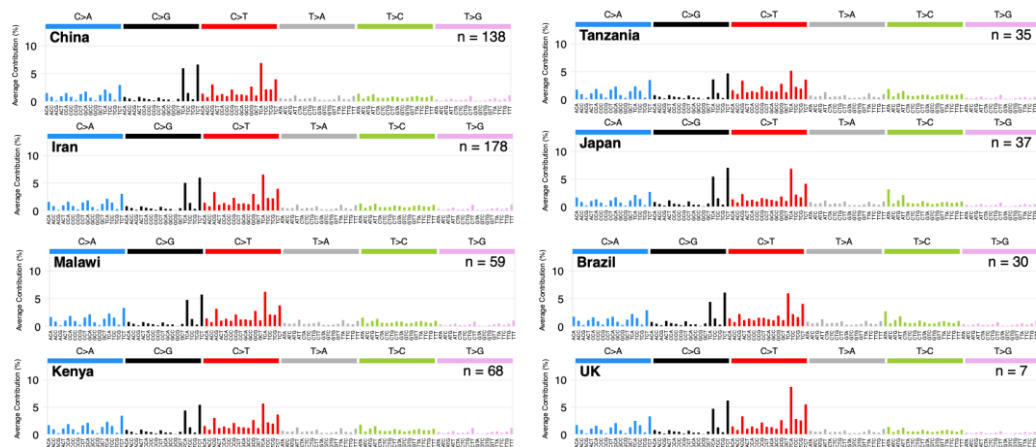


ESCC Mutographs collection

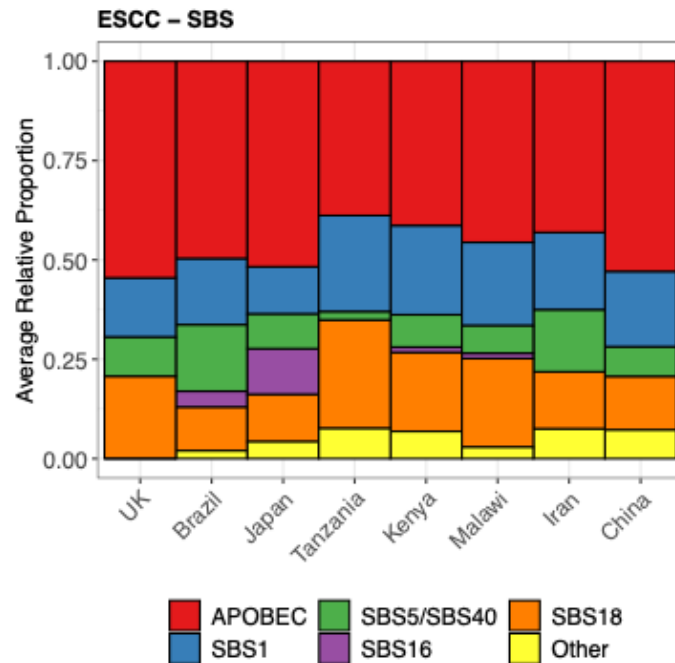


ESCC has very high incidence rates in parts of Asia and East Africa
The Mutographs collection included 552 cases from 8 countries

Esophageal Squamous Cell Carcinoma (ESCC)



Remarkably, the mutational signature landscape of ESCC was remarkably similar across all countries



Mouse Signatures

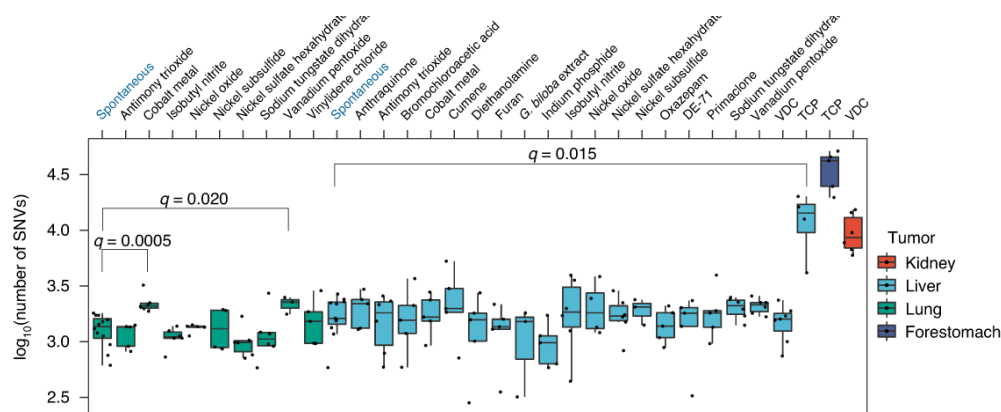
Article | Published: 28 September 2020

The mutational signature profile of known and suspected human carcinogens in mice

[Laura Riva](#), [Arun R. Pandiri](#), [Yun Rose Li](#), [Alastair Droop](#), [James Hewinson](#), [Michael A. Quail](#), [Vivek Iyer](#), [Rebecca Shepherd](#), [Ronald A. Herbert](#), [Peter J. Campbell](#), [Robert C. Sills](#), [Ludmil B. Alexandrov](#), [Allan Balmain](#) ✉ & [David J. Adams](#) ✉

Nature Genetics 52, 1189–1197 (2020) | [Cite this article](#)

18k Accesses | 150 Citations | 183 Altmetric | [Metrics](#)

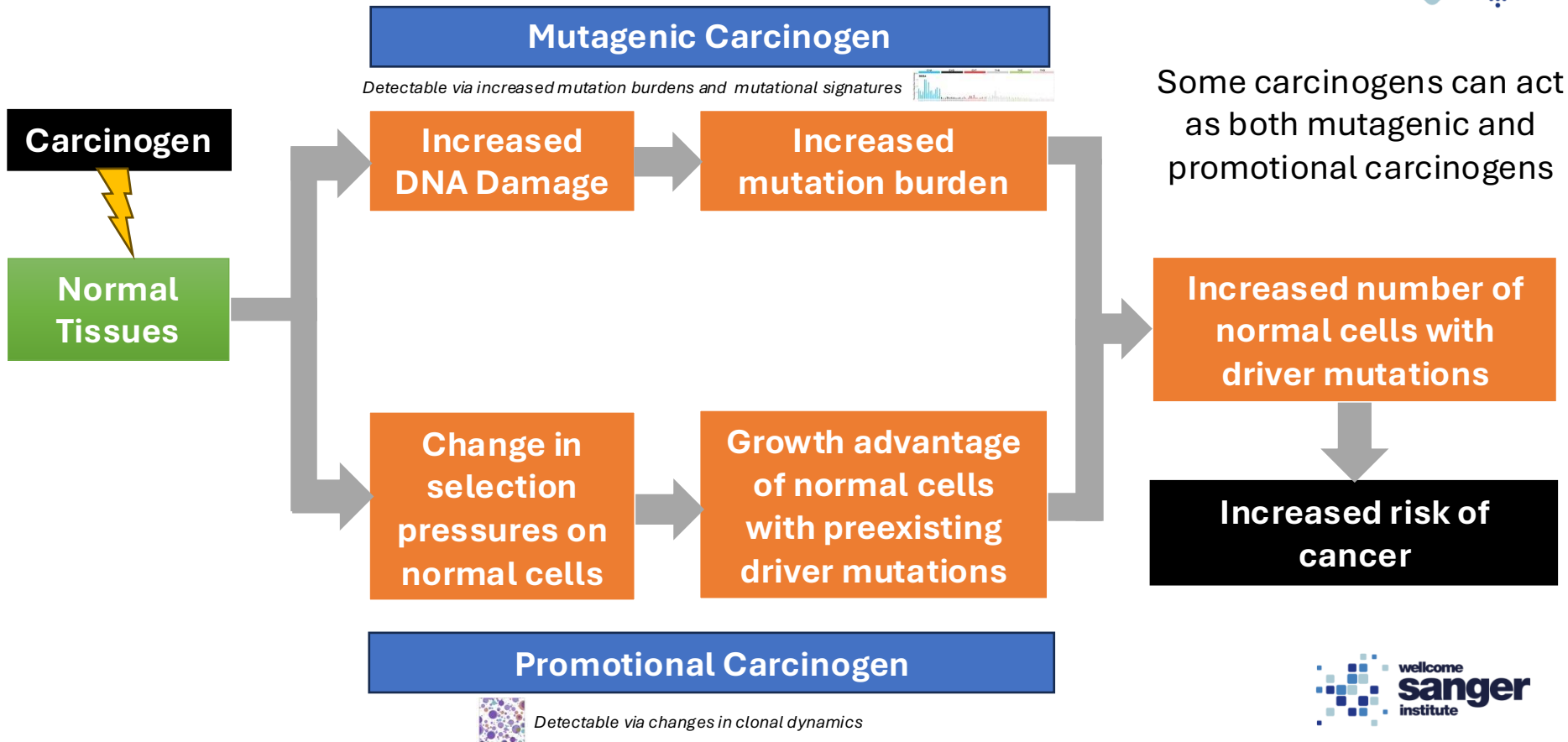


This finding coincided with other work within Mutographs looking at carcinogens in mice

Surprisingly, only a small number of the tested compounds generated mutational signatures

For most, mutation burden was comparable to those of spontaneous tumours

Promotional vs Mutagenic Carcinogens



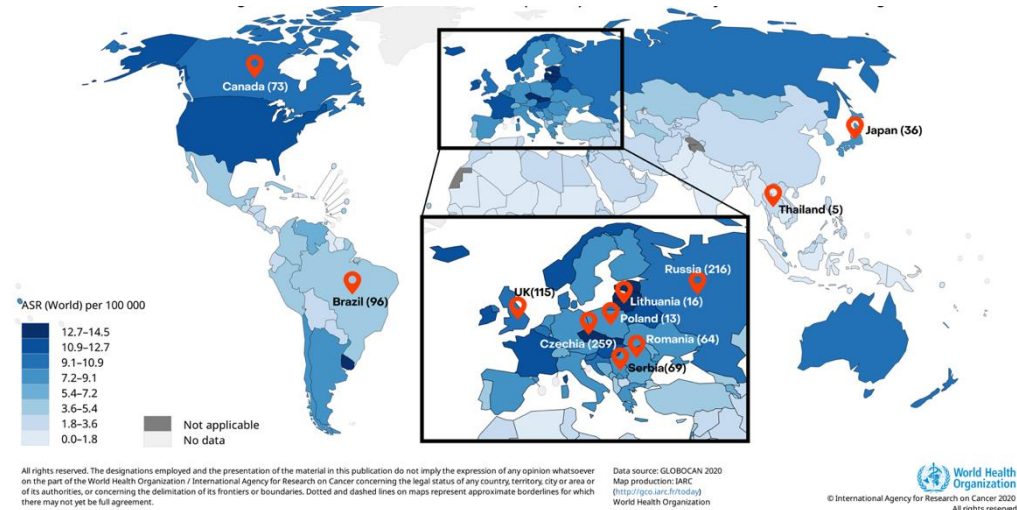
Kidney Cancer

The Mutographs cohort included 962 cases of clear cell renal cell carcinoma (CCRCC)

Cases were collected from 11 countries ranging from low to high incidence rate

Known or suspected CCRCC risk factors include:

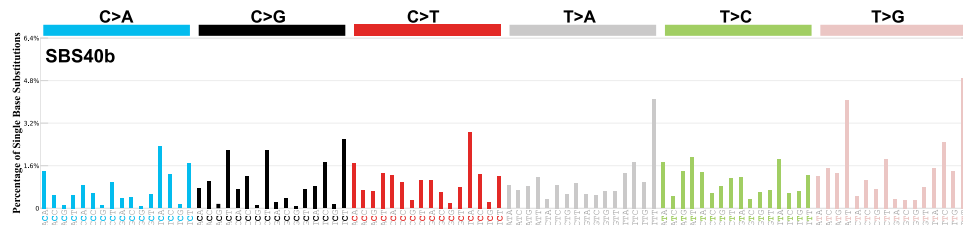
- Hypertension
- Obesity
- Tobacco Smoking
- Age
- Sex
- Family History
- Diabetes
- PFAS (per/polyfluoroalkyl substances)



3 key findings emerged from this study

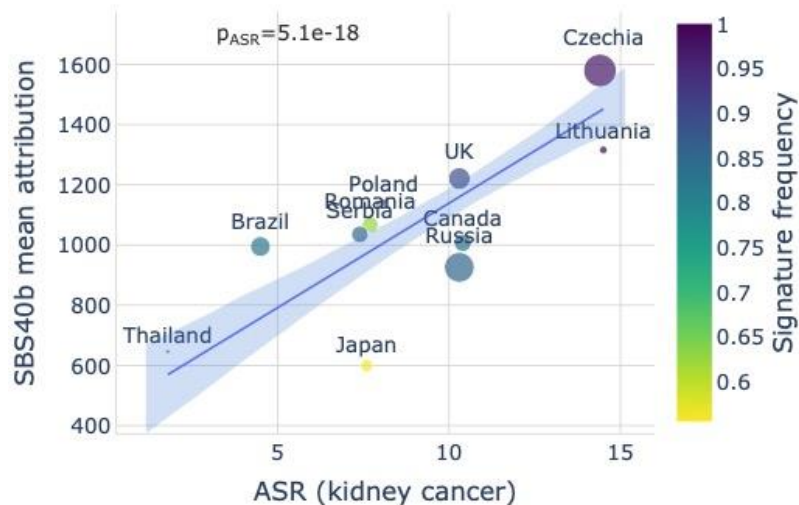
Kidney Cancer

Key Finding 1



SBS40b associates with the age standardised rate (ASR) of kidney cancer

ASR/SBS40b association across countries



SBS40b is associated with age, and also correlated with both ID5 and ID8 which are also clock like signatures.

SBS40b mutation burden is positively correlated with biomarkers of impaired kidney function

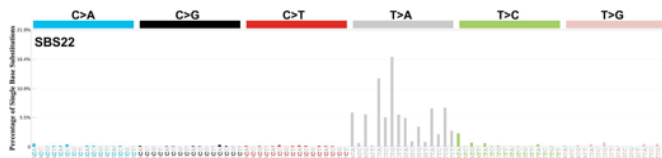
Kidney Cancer

Key Finding 2

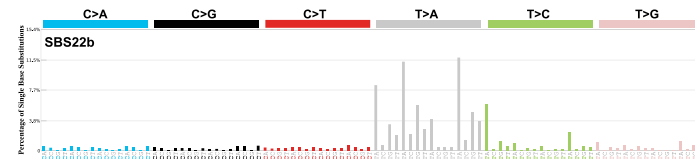
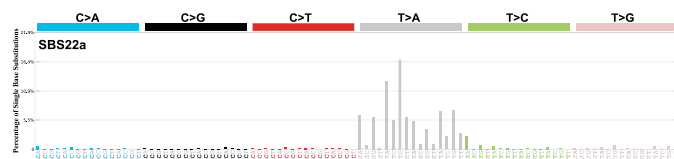
Aristolochic Acid is found in extracts of plants in the *Aristolochia* genus and is known to be mutagenic (SBS22). It was previously found in RCC from Romania



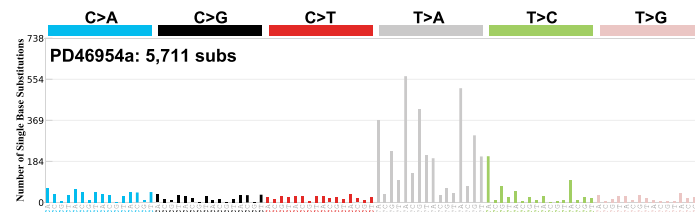
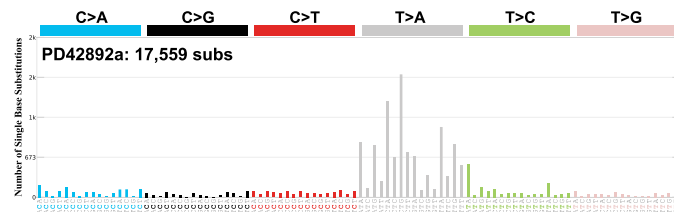
Reference
Signature



Mutographs
AA related
Signatures

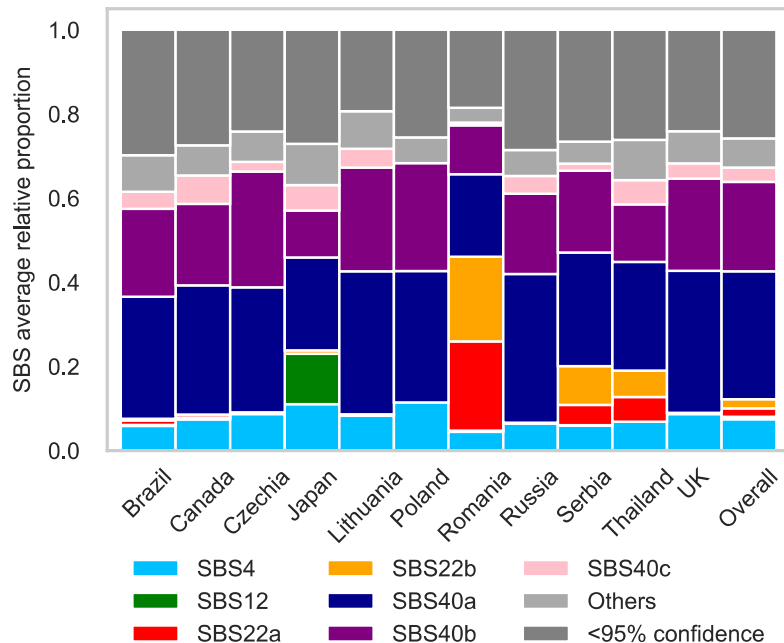


Representative
cases



Kidney Cancer

Key Finding 2



Aristolochic Acid Signatures can be found not just in Romania but also in Serbia (and Thailand)

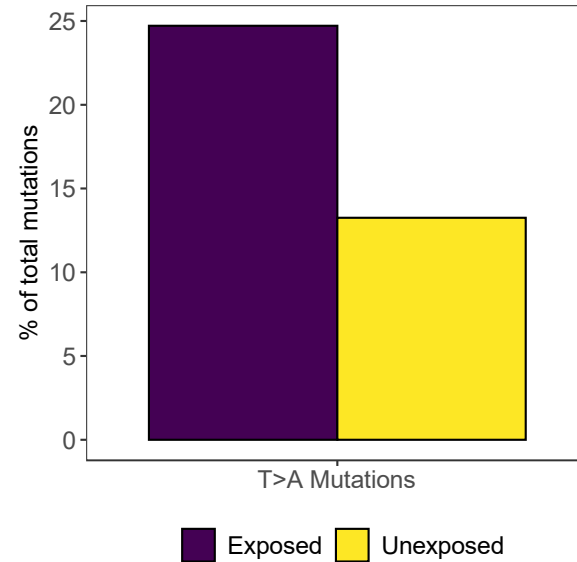
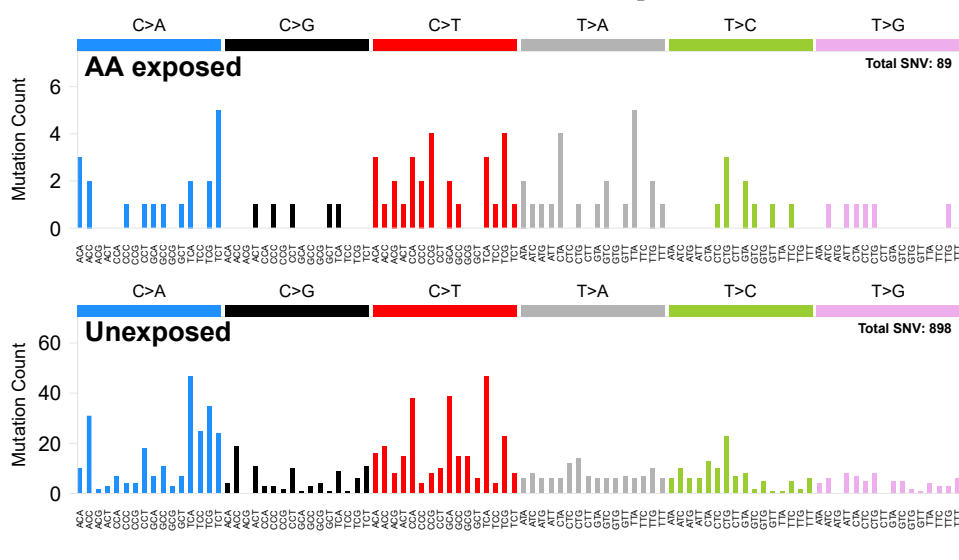
This suggest that potentially millions of individuals in South-Eastern Europe are being exposed

A follow-up study is in progress to determine the extent of exposure across the Balkan region

Kidney Cancer

Key Finding 2

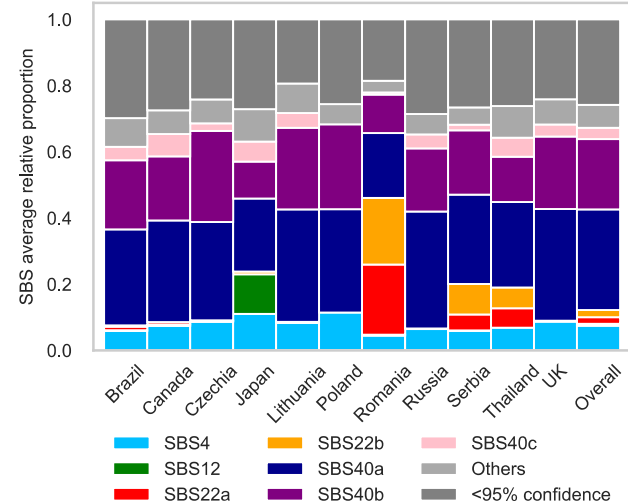
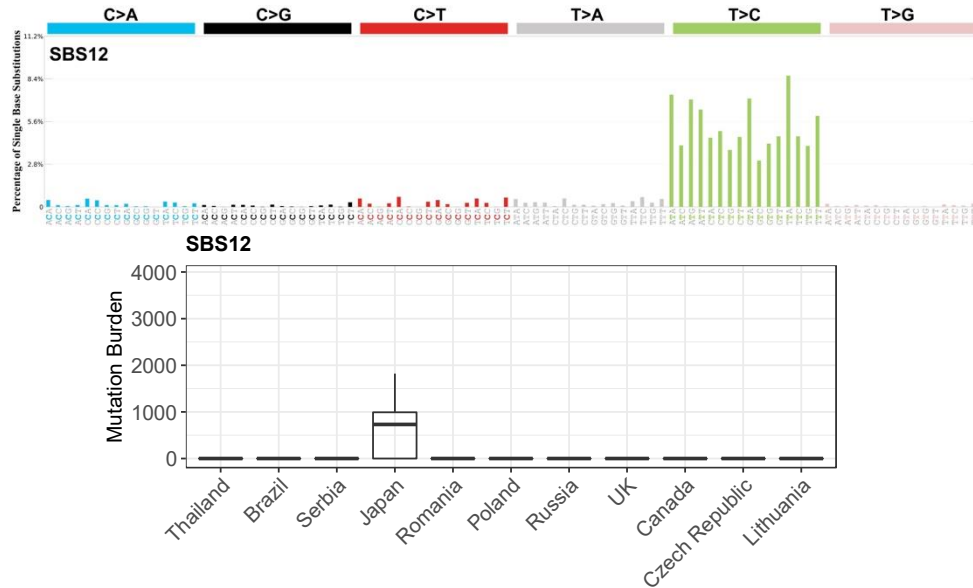
Driver Mutation Spectra



There is a significant difference in the number of T>A mutations classified as likely driver mutations in AA exposed cases (p-value=0.0062)

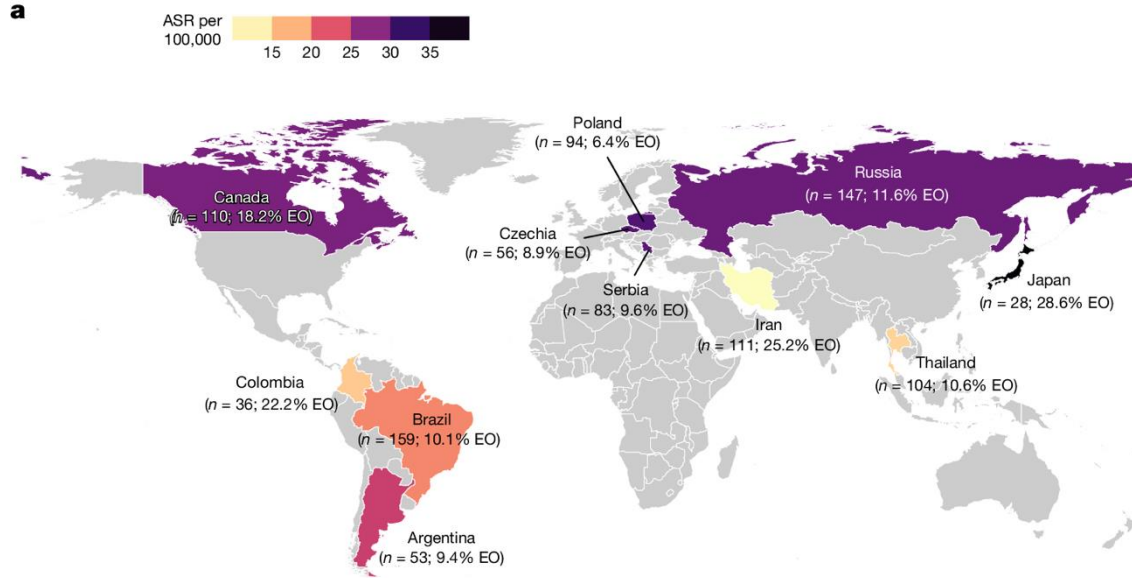
Kidney Cancer

Key Finding 3



SBS12 is present in >70% of Japanese cases but very rare (<2%) elsewhere
 No association with RCC risk factors was found
 SBS12 shows features typical of an exogenous exposure

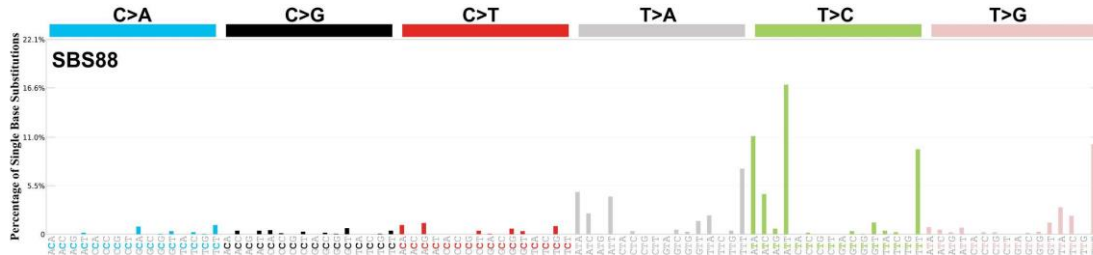
Colorectal Cancer



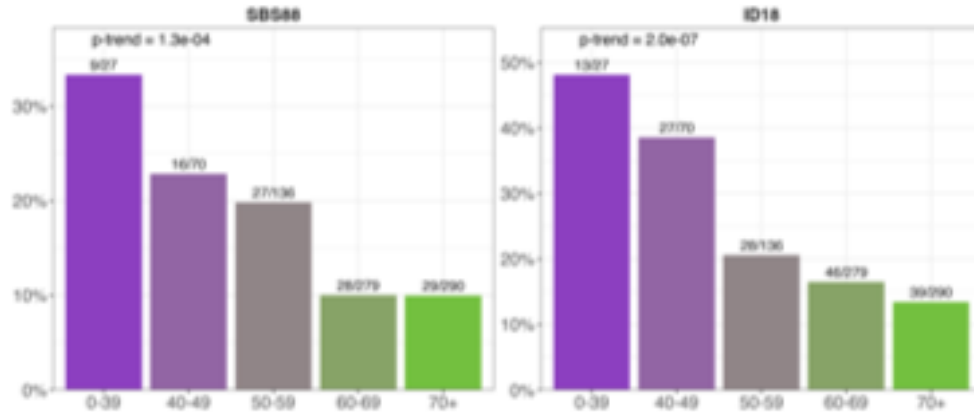
The Mutographs cohort included 981 cases of colorectal cancer from 11 countries

Key to this study was investigating the rise of early onset individuals (below 50 years of age) which has doubled in the past 20 years

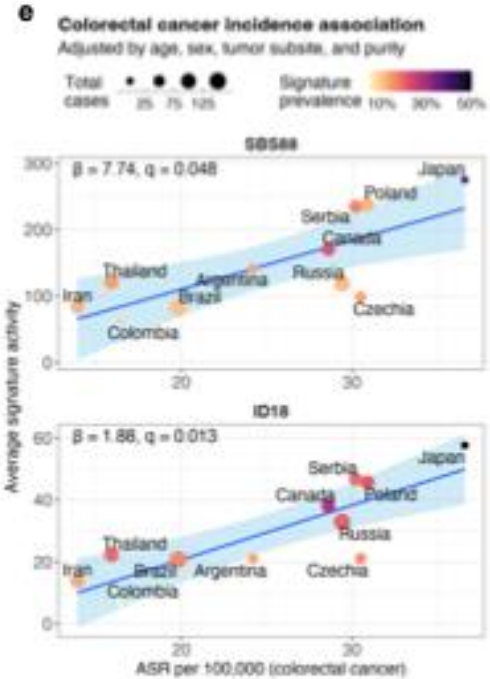
Colorectal Cancer



C Age of onset trend enrichment
Adjusted by sex, country, tumor subsite, and purity



SBS88 is enriched in early onset colorectal cancer and associates with age standardized incidence rate





Summary – Part 1

The study of somatic mutations and mutational signatures are a powerful tool for investigating the causes of cancer

However, the mutational epidemiology approach used by Mutographs has some limitations:

- Promotional carcinogens are largely invisible
- If a signature is detected, it can be very difficult to establish a hypothesis for the causative agent